Exploratory analysis after cleaninh

FAROUK OSMAN

2026-11-03

INTRODUCTION

This is an exploratory analysis of a raw medical dataset that was posted on Kaggle.

The dataset contains 100 patient records with 10 variables capturing demographic characteristics, clinical measurements and diagnostic outcomes. Following a data cleaning exercise conducted in R, the cleaned dataset is used here to explore patterns in clinical characteristics and their relationship to patient diagnosis.

The analysis is guided by three objectives: to describe the distribution of key clinical characteristics across diagnosis groups; to examine associations between patient characteristics and Heart Disease; and to identify the strongest independent predictors of Heart Disease using logistic regression.

This dataset is synthetic; useful for learning purposes and findings cannot be generalised to real clinical populations.